London Ambulance Service NHS
NHS Trust

Support the management of trauma in the emergency and urgent care setting

Component 8

Head Injuries

Head Injuries - Component 8



Objectives

- Describe mechanisms associated with head injuries
- Describe the signs, symptoms and complications of head injuries
- Explain the treatment for head injuries and when intervention from a senior clinician may be required

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Head Injuries

Common causes of traumatic brain injury (TBI) include:

- Road traffic collisions (RTCs)
- Falls
- Assault
- Direct blows to the head



The worst outcomes are associated with:

- Penetrating injuries
- Non-accidental injury in children <5 years
- Pedestrians & pedal cyclists
- Ejection from a vehicle



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Head Injuries



Primary Brain Injury:

Instantaneous injury resulting from the impact itself.

Once a primary brain injury has occurred, it cannot be undone, i.e. primary injury cannot be treated – the damage has already been done.

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Head Injuries

Good pre-hospital management can help influence the impact of secondary brain injury & involves:

- Recognising that a traumatic brain injury (TBI) may be present.
- Understanding the pathophysiology relating to TBI.
- Managing the patient appropriately.

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Traumatic Brain Injury (TBI)

TBI is “Brain injury caused by trauma to the head”



Complications of brain injury include:

- Hypoxaemia
- Hypotension
- Intracerebral Haematoma

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Recognition

Suggestive symptoms of TBI include:



- Lacerations, contusions or haematomas to scalp
- Boggy areas
- Visible fractures/deformity
- Battle's sign or 'panda eyes'
- Cerebrospinal fluid & or blood leaking from nose & ears
- Cushing's Triad - raising blood pressure, reducing heart rate, & irregular respirations
- Dizziness
- Nausea &vomiting
- Abnormal pupils & or reaction

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Recognition

Suggestive symptoms of TBI include:



- Visual disturbances
- Severe headache
- Altered level of consciousness
- Repetitiveness
- Amnesia
- Paraesthesia/paralysis of extremities
- Convulsions
- Posturing
- Abnormal respirations

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Recognition – scalp lacerations

- Scalp is a very vascular area
- Injuries may appear worse than they are and distract you from underlying/other injuries that may be present

Generally not severe enough to cause Hypovolaemic shock, **except:**

- If an artery has been damaged blood loss can be significant
- Scalp injuries in children can cause hypovolaemia

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Recognition - skull fractures

- Injury to rigid box around brain and indicates significant force
- May or may not be associated with scalp injury
- Common complication of skull fractures are damage to brain structures underneath

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Recognition

Periorbital ecchymosis

(Pandas Eyes)



Bruising behind the ear

(mastoid process) – Battles Sign



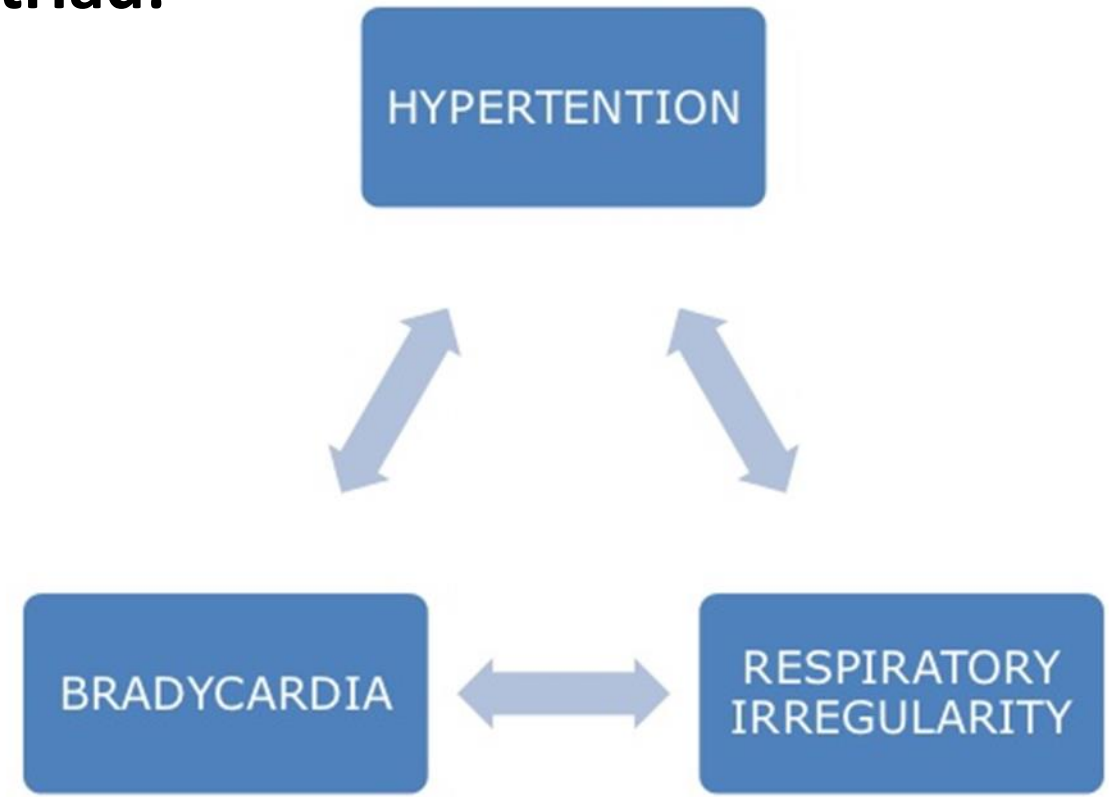
Not apparent for several hours!

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Recognition

Cushing's triad:



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Recognition

- Frequently monitor the size, equality and reactivity of pupils.
- Nerves that control dilation and constriction are very sensitive to ICP.
- A pupil that is 'sluggish' (slow) to react is an early sign of ICP. It may also indicate cerebral hypoxia.
- Unequal or bilaterally fixed and dilated (blown) pupils are later more ominous signs of ICP and indicate pressure on the oculomotor nerves





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Recognition

Altered level of consciousness (LOC):

The best indicator of a patient's condition is their level of consciousness

- AVPU System
- Glasgow Coma Scale (GCS)
- Pupillary Response

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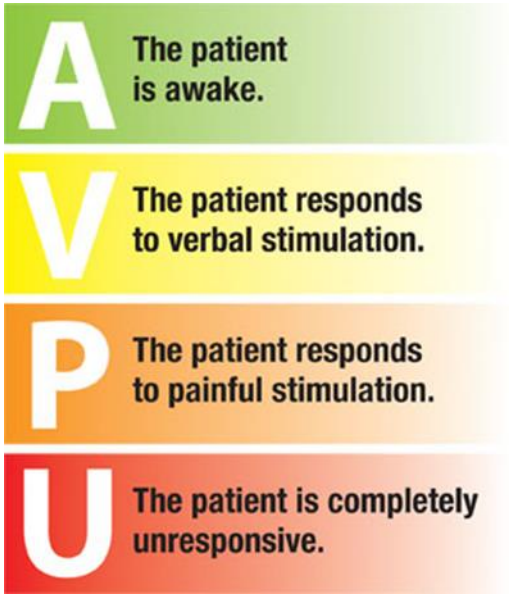
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Recognition

AVPU Scale:

Baseline neurological assessment indicates basic level of consciousness.

It is not a substitute for calculating the full GCS.



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Recognition

Glasgow Coma Scale (GCS):

- A neurological scale that aims to give a reliable, objective way of recording the conscious state of a person for initial as well as subsequent assessment.
- Based on three measurements: eye opening, verbal response and motor response.
- Used to determine the severity of brain injury, it can also be used in the acute medical or trauma patient.
- A patient is assessed against the criteria of the scale, and the resulting points give a patient score between 3 and 15, the lower the score, the more severe the extent of brain injury.

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Recognition

GCS: The most important single sign in a head injured patient is a changing state of consciousness.

Glasgow coma scale		Score
Eye opening	spontaneously to speech to pain none	4 3 2 1
Verbal response	orientated confused inappropriate incomprehensible none	5 4 3 2 1
Motor response	obeys commands localises to pain withdraws from pain flexion to pain extension to pain none	6 5 4 3 2 1
Maximum score		15

Brain injury classification based on GCS:
13-15 =mild traumatic brain injury
8-12 = moderate traumatic brain injury
3-8 = severe traumatic brain injury

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Trapezius Squeeze



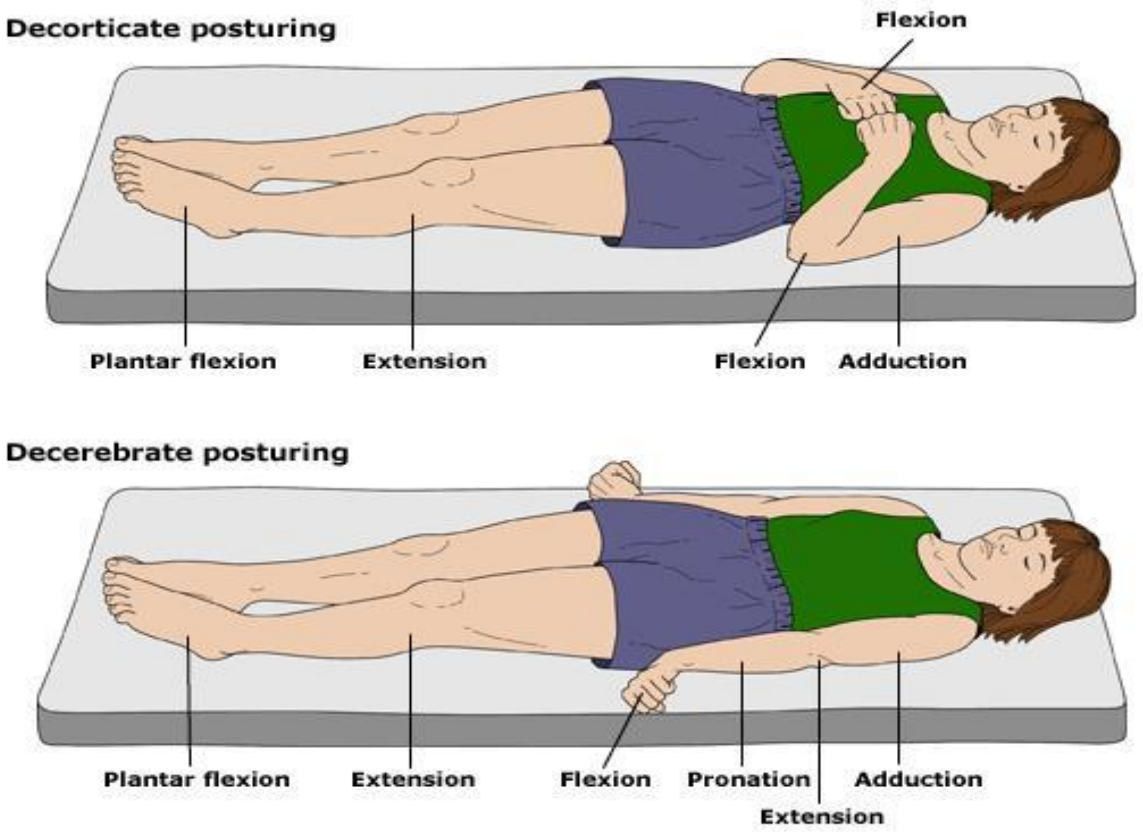
Supra-orbital pressure

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Recognition

Posturing:

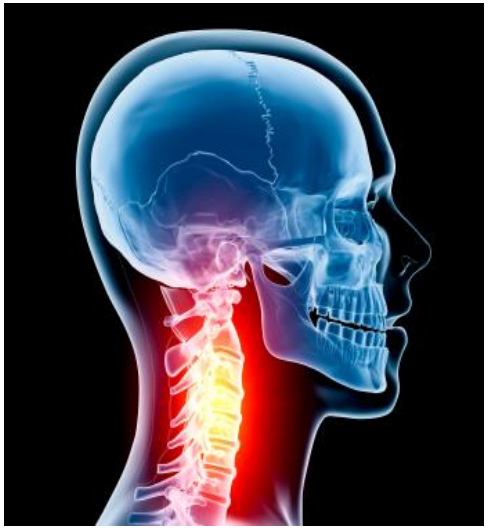


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Complications

- Head injuries are often complicated by cervical spine injuries
- There is significant association between reduced levels of consciousness (LOC) and C-spine injury
- Maybe easily overlooked. Always assume a C-spine injury.
- Protect the C-spine to prevent further damage



SIGNIFICANT HEAD INJURY PATIENTS = FULL IMMOBILISATION

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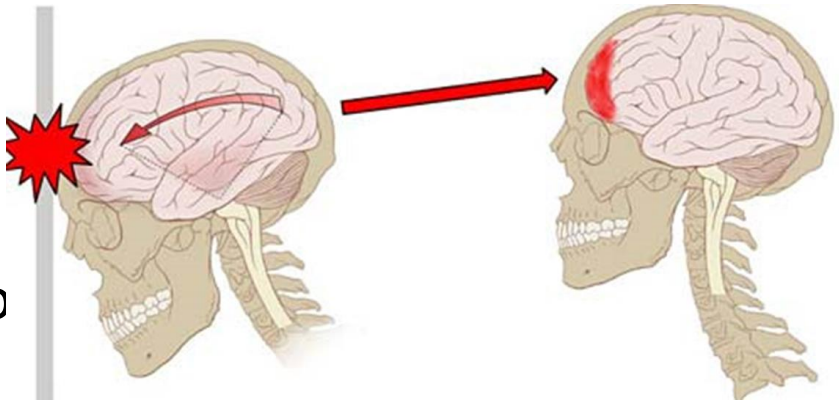
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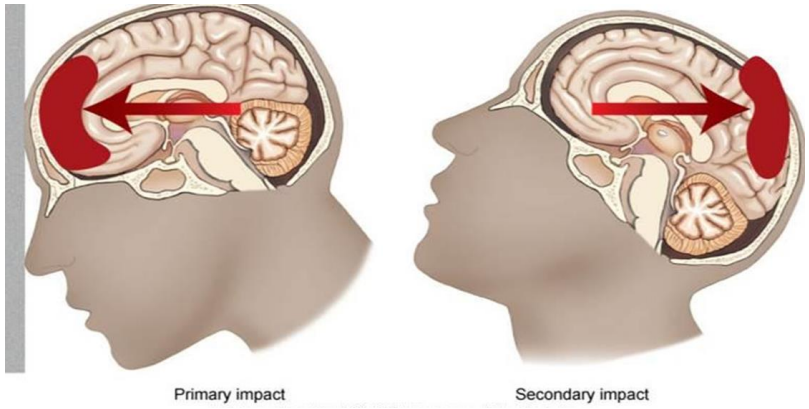
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Complications

Cerebral contusion:

- Bruising, swelling
- Results from brain hitting skull's inside causing physical damage to brain tissue
- Coup-/contrecoup pattern
- Since brain is in closed box, pressure increases as brain swells, blood flow to brain **decreases.**





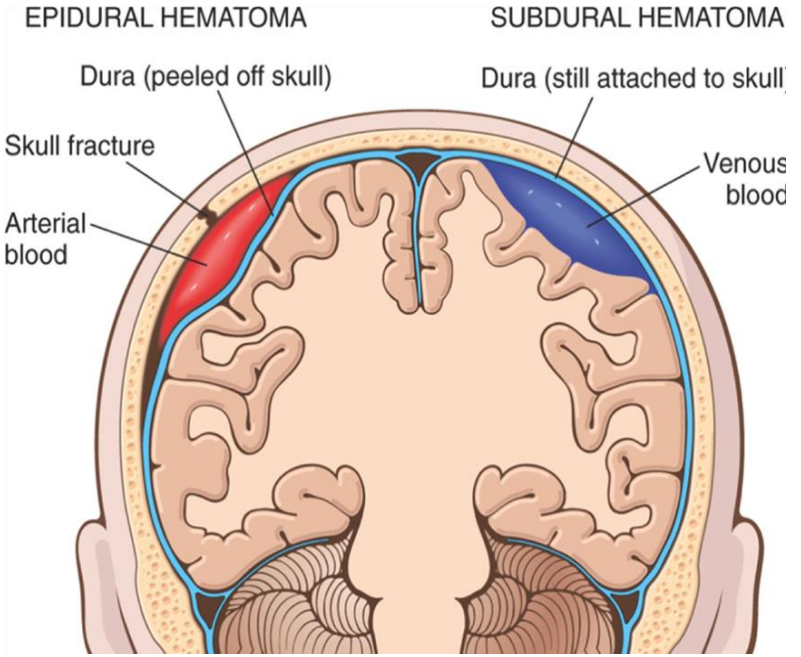
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Complications

Intracranial bleeding is a complication of TBI. Haemorrhage occurs in different parts of the brain such as:

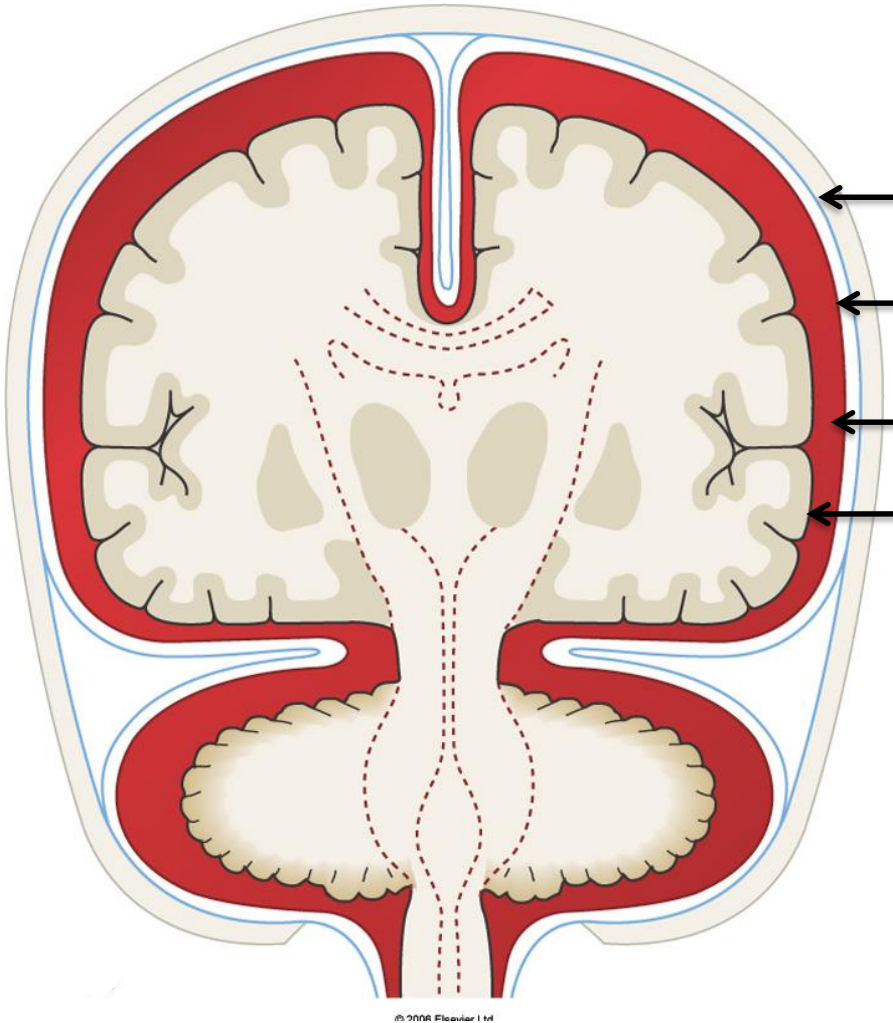
- Extradural Haemorrhage
- Sub Dural Haemorrhage
  - Acute
  - Chronic
- Sub arachnoid Haemorrhage



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Sub Arachnoid Haemorrhage



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Pathophysiology

In the traumatically injured brain:

- Autoregulation starts to fail
- Blood flow in affected areas of the brain changes
- It relies on higher than normal mean arterial pressures (MAP) to maintain perfusion

Body responds to increasing intracranial pressure by:

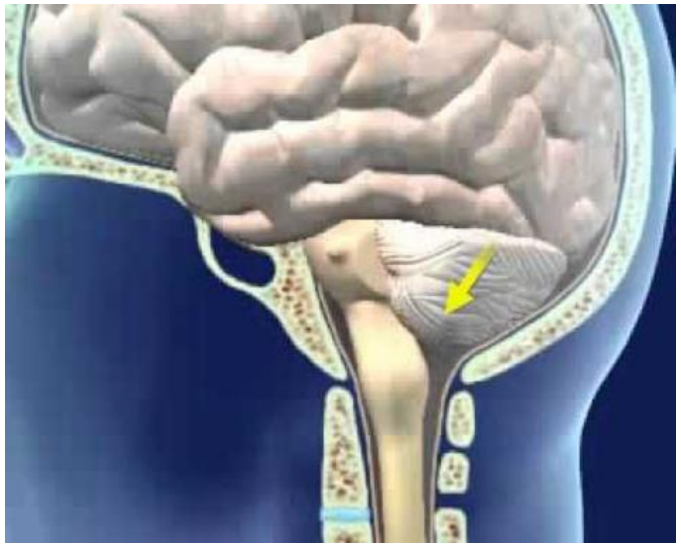
- Raising BP
- Increased BP, moves blood into brain against rising ICP
- Heart rate falls in response to rising BP

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Pathophysiology

- Early detection of **increased intracranial pressure (IPC)** is critical.
- If pressure inside skull exceeds average blood pressure, blood flow to brain stops.
- Increasing intracranial pressure can force brain downward into spinal canal, crushing it.



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Management



Pre hospital aims: To ensure adequate perfusion to the brain by:

- Prevent hypoxia
- Treat hypotension
- Maintain EtCO2 (normocapnia)
- Transport patient(s) quickly for definitive care

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Management

Senior Intervention:

- Serious head injured patients require specialist input early – Request support early
- **Paramedic** – Can assist with advanced airway management procedures
- **Advanced Paramedic Practitioner** – As paramedic plus drugs to sedate patient if needed (combative/agitated)
- **BASICS/HEMS** - May anaesthetise the patient to protect the airway and ensure adequate ventilation, can also perform other specialist skills and interventions

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Management

- Do **NOT** apply pressure to open or depressed skull fractures
- Do **NOT** attempt to stop flow of blood or CSF from nose or ears
- Do **NOT** remove penetrating objects.
- Do **NOT** spend extended time on scene
- **Do NOT forget the C-spine!!!!**

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Management

Vital signs:

Isolated head injury does not cause hypotension or tachycardia!

Signs of shock in head injured patient indicate other injuries are present!

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Table 4.11 – CONVEYANCE DECISION TOOL

Red criteria

Refer to local major trauma triage tool and consider the need for immediate transfer to a major trauma centre and/or pre-alert call.

Immediately transport to hospital if any one of the following is found:

- Glasgow coma scale (GCS) score of less than 15 on initial assessment.
- Any loss of consciousness as a result of the injury.
- Any focal neurological deficit since the injury.
- Any suspicion of a skull fracture or penetrating head injury since the injury.
- Amnesia for events before or after the injury.
- Persistent headache since the injury.
- Any vomiting episodes since the injury (clinical judgement should be used regarding the cause of vomiting in those aged 12 years or younger and the need for referral).
- Any seizure since the injury.
- Any previous brain surgery.
- A high-energy head injury.
- Any history of bleeding or clotting disorders.
- Current anticoagulant therapy such as warfarin (see further detail below).
- Current drug or alcohol intoxication.
- Any safeguarding concerns (for example, possible non-accidental injury or a vulnerable person is affected).
- Continuing concern by the ambulance clinician about the diagnosis.
- Irritability or altered behaviour, particularly in infants and children aged under 5 years.
- No visible trauma to the head but still of concern to the clinician.
- No one is able to observe the injured person at home.
- Continuing concern by the injured person or their family or carer about the diagnosis.

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NICE guidance for head injury patients on anti-coagulants

Section 1.4.12

Patients having warfarin treatment

1.4.12 For patients (adults and children) who have sustained a head injury with no other indications for a CT head scan and who are having warfarin treatment, perform a CT head scan within 8 hours of the injury. A provisional written radiology report should be made available within 1 hour of the scan being performed.

Therefore head-injured patients on Warfarin should be conveyed to an ED regardless of apparent severity

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Summary

- Remember importance of essential airway management – using adjuncts if needed
- Consider assisting ventilations if RR <10 or >30/min
- Patients could be combative/aggressive and challenging to manage – request senior input early
- Accurately assess disability using AVPU,GCS and pupil assessment and reassess regularly to recognise deterioration
- Differentiate between Major and Minor

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Any Questions?





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